Full system overview

Phone-based vision processing. The main ESP32-S3 owns helmet warnings, suspected-crash logic, SOS and emergency communication.

FRONT CAMERA NODE

ESP32-S3-CAM + camera
Camera driver + PSRAM buffers
Frame ID / capture timestamp
JPEG capture + Wi-Fi streaming
Exposure / image-health reports

REAR CAMERA NODE

ESP32-S3-CAM + camera
JPEG capture + timestamped frames
PSRAM buffers + Wi-Fi streaming
Mounting / view calibration
Placement decision: bike-fixed for stable zones; helmet view needs pose validity.

JPEG

JPEG

PHONE APPLICATION

Stream and timing manager

Decode newest frames; per-camera queues, clock mapping and health.

Pothole model

Road image region
Detection + persistence
Quality / age gating

Wetness model

Dry / damp / wet
Standing water / unknown
Stable classification

Rear vehicle pipeline

Detect + track vehicles; calibrated zones; optional camera distance estimate; pose and coverage validity.

Outbound: current observations + validity + frame age. No direct motor control.

Results

Events

MAIN HELMET CONTROLLER

ESP32-S3 N16R8

Sensor acquisition and input states
Motion / wear / eye / vapour
Suspected-crash state machine
Candidate / verify / countdown / SOS
Phone result validation
Session / sequence / freshness
Central warning manager
Priority + left/right output validity
Independent workers
GNSS / LTE / SD / health

Proposed network: main board hosts the Wi-Fi access point. Phone and camera boards join it. Video uses the AP radio; benchmark the combined load.

Local control

LOCAL INPUTS AND SENSOR INTERFACES

MPU6050: I2C + interrupt | VL53L0X: separate I2C | Strap switch: digital
OP0062: conditioned digital input | MQ-3B: scaled ADC1 input
Cancel + SOS: digital inputs | GNSS + antenna: UART | Battery monitor: ADC1

CONTROLLED OUTPUTS

Two haptic motors + drivers; buzzer; LEDs
LTE + antenna + SIM: emergency SMS
microSD + internal flash: event records

Detailed maps: controller (02), phone AI (03), incident flow (04), power and faults (05), evidence and expansion (06).

POWER: protected source + charging / power path + regulated branches

Main ESP32-S3: modules and submodules

Separate bounded workers keep motion sampling and incident deadlines independent of networking, storage and modem operations.

MPU6050
Acceleration + rotation
Data ready / clipping / sample gaps

VL53L0X + strap switch
Head presence + strap position
Present / absent / unknown

OP0062
Polarity / level conditioning
Open / closed input + calibration

MQ-3B
Heater readiness + baseline
Scaled analog vapour response

Cancel + SOS buttons
Debounce / deliberate hold
Reject held-at-entry cancellation

GNSS + battery monitor
Fix quality / position / fix age
Battery voltage / supply state

Motion

INPUT BUS

Acquisition and input drivers
IMU interrupt worker; bounded I2C / UART; ADC scaling; switch debounce; timestamp + validity on every observation.

Validated observations

Wear, eye and pre-ride state modules
Wear: head + strap truth table; recent wear history
Eye: closure timer; blink rejection; unavailable state
Vapour: conditioning, baseline, sampling and invalid state
Wear / eye / vapour publish directly to the warning manager.

Recent wear

Motion features and incident state machine
Peak / duration / rotation / stillness + clipping flags
Candidate verification + recent wear context
Countdown, new cancel action, deliberate SOS and incident ID

Incident status

Warning manager - sole output owner
Fresh vision + local sensor warnings + incident status
Select urgency / side / pattern; expire stale advisories
Emergency sequence has priority over advisory outputs.

Scheduler, configuration and health
Bounded queues; progress watchdog; reset cause; calibration and contact settings; battery / degraded modes.

Incident

Emergency worker
GNSS fix selection
LTE AT commands + deadlines
Registration / send / retry backoff
Submitted vs delivery-reported
Phone failure does not block SOS.

Storage worker
PSRAM/RAM rolling history
microSD append + checksum
Internal-flash pending-event journal
Recovery / full / missing handling

Output driver module
Left haptic + right haptic drivers
Buzzer driver + status LEDs
Non-blocking pulse scheduling
Measured actuator / IMU interaction

Phone / network adapter
Wi-Fi AP + paired app session
WebSocket observations / events
Schema + source + sequence checks
Capture-age mapping + expiry
Separate stream / inference health
No vision-triggered emergency SMS

Exact GPIOs and voltage levels depend on the selected breakout boards.

Camera and phone AI: complete processing path

The camera nodes capture frames. The phone runs inference and tracking. All audible and haptic alerts are produced by the main board.

Front node: capture and stream

Camera driver -> PSRAM frame buffer -> timestamp / frame ID -> JPEG publisher

Wi-Fi video

Phone: front stream and quality gate

Decode newest frame; map capture time; check blur / exposure / occlusion; maintain stream and inference health.

Pothole submodules

Road image-region selection
Detector + confidence filtering
Temporal association
Persistence / quality gate
Visible pothole observations

Wetness submodules

Road-region crop / mask
Dry / damp / wet classifier
Standing-water class
Unknown / out-of-domain state
Temporal smoothing

Front observation state

Pothole visibility + image region; wetness class + validity. Image region does not establish the motorcycle travel path.

Rear node: capture and stream

Camera driver -> PSRAM frame buffer -> timestamp / frame ID -> JPEG publisher

Wi-Fi video

Phone: rear stream and quality gate

Decode newest frame; map capture time; check image quality, frame gaps, camera pose and usable coverage.

Vehicle detector

Cars / motorcycles / bicycles / buses / trucks; evaluate local three-wheelers. Associate detections with track IDs.

Tracking and zone logic

Track age / continuity
Occupancy / approach cues
Motorcycle-relative zones only when geometry is valid
Lost coverage -> unknown

Distance estimate

Calibrated camera intrinsics
Known-size or road-plane model
Estimate + uncertainty / validity
No metric result when invalid
Radar is an optional input.

Local phone modules: settings, model versions, latency metrics, debug view, short evidence buffer.

Observation publisher -> main ESP32-S3

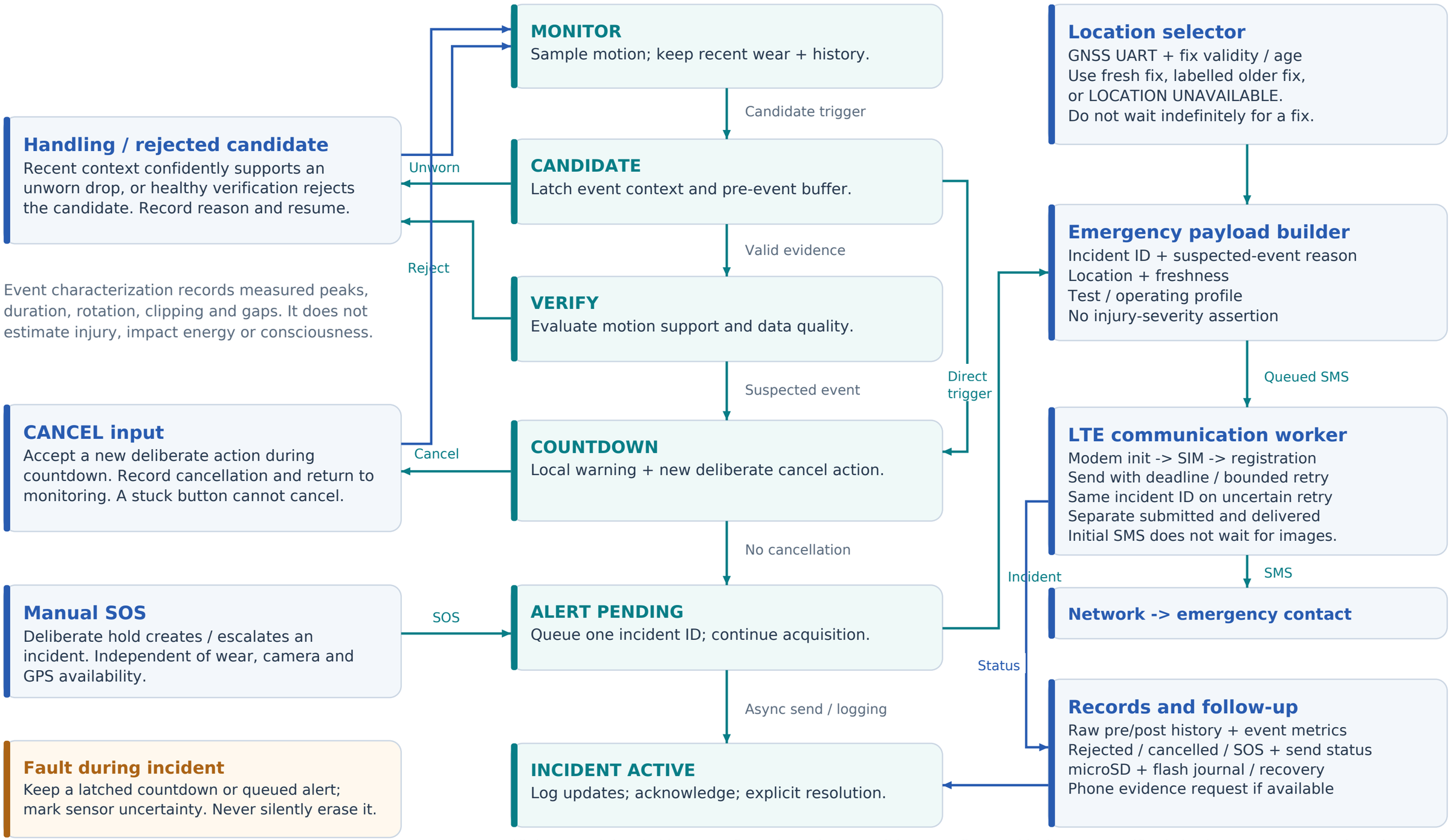
Session / sequence / frame ID / capture time / clock uncertainty / health / validity / class / track / zone / optional distance estimate. Bounded messages and queues.

Main board returns: incident requests, settings, time-sync exchanges and connection status.

Degraded state is explicit. A live app connection does not prove that recent usable camera frames or recent inference results exist.

Incident detection and emergency workflow

Motion and deliberate SOS create incidents. Camera detections are advisory inputs and never independently trigger emergency messages.



Event characterization records measured peaks, duration, rotation, clipping and gaps. It does not estimate injury, impact energy or consciousness.

All timers and thresholds are configuration values to validate on a fixture; this diagram specifies behaviour, not proven road-crash sensitivity.

Power, interfaces and fault isolation

Rail values and current ratings remain board-specific. Select regulators and protection after measuring simultaneous camera, LTE, heater and haptic loads.

Power inputs
Protected battery pack
Charging / bench USB input
Source and charge ratings: to select

Protection and power path
Charger + suitable power-path control
Protection / fuse / main switch
Input monitoring and load policy

Regulated distribution
Separately rated branches
Bulk and local decoupling
Grounding + connectors + wiring

Main MCU + sensors
ESP32-S3 N16R8
MPU6050 / VL53L0X / GNSS
Conditioned OP0062 / switches
microSD and logic interfaces

LTE branch
Exact module input voltage
Transmit-current headroom
Antenna + SIM
UART level compatibility

Camera branches
Front + rear ESP32-S3-CAM
PSRAM / camera supplies
Load control if provided
Network load / heat testing

Heater + actuators
MQ-3B heater supply
Left / right motor drivers
Buzzer and LEDs
Inductive-load protection

Health and power supervisor
Battery ADC + calibrated thresholds
Task progress / watchdog / reset cause
IMU / SD / LTE / camera health flags
Low power: preserve incident work and reduce lower-priority loads when supported.

Data and recovery boundaries
Bounded RAM queues count dropped data
microSD errors become LOG_DEGRADED
Flash journal recovers complete records
Shared battery / main MCU remain single failure points; no guaranteed crash survival.

Electrical and mechanical checks
Validate input levels / pull-ups / ADC scaling
Do not connect motors to GPIO power
Verify peak load, heating and resets
Fixture mounting first; no shell / impact-liner cutting or drilling.

DEGRADED-MODE CONTRACT
Camera / phone lost: vision unavailable; crash and SOS continue. **GNSS missing:** labelled old fix or no location.
LTE unavailable: local warning and bounded retries. **SD failed:** continue detection and sending; record data loss.
IMU invalid: automatic motion detection unavailable; deliberate SOS and already-latched incident work remain active.

Evidence, configuration and expansion ports

Solid blocks are in the proposed architecture. Dashed orange blocks are uncommitted extensions; they do not establish that hardware or software exists.

EVENT EVIDENCE AND CONFIGURATION

Main event request
Incident ID + request time
Bounded request to phone
Initial SMS proceeds independently.

Phone image-evidence module
Short front/rear frame buffers; preserve available stills on event request; store capture time, frame age, source and quality.

Local incident records
Main: microSD + flash journal
Phone: images + event metadata
Link using the same incident ID
Retention / deletion settings

Image sharing - conditional
If a selected data connection and endpoint are available: upload over HTTPS, then send an evidence link. SMS cannot contain an image attachment.

Settings and pairing
Device / session identities + credentials
Emergency test contacts
Sensor / camera calibration profiles
Warning configuration + validity limits
Model and firmware version records

Offline model-development path
Capture / label representative data
Split by road / capture session
Train -> export -> phone deployment
Profile accuracy + end-to-end latency
Record model / dataset / config versions

OPTIONAL: radar range input
HLK-LD2451 or another evaluated vehicle radar -> bike node -> wireless target reports. Associate radar targets with camera tracks; validate range, angle, coverage and closing speed.

PROPOSED: rider-intent input
Left / right demo buttons first. A later bike interface can report turn signals using suitable isolation / level protection. Head movement alone is not lane-change intent.

FUTURE: display / update interfaces
Optional display or HUD consumes main-board warning state. Future authenticated firmware/model updates require version checks, recovery and an explicit update mode.

INTEGRATION AND ACCEPTANCE MODULES

Sensor fixtures / simulated inputs clearly labelled; real test SMS; fresh/stale/no-fix cases; cancel / SOS; SD failure; phone disconnect; dropped frames; both AI streams; motor/IMU interaction; cold boot and power-load tests.
Evidence: missed events, wrong-side alerts, nuisance rate, capture-to-actuator delay, unavailable time and reset / data-loss logs.

OPEN BUILD DECISIONS

Exact phone / camera / LTE boards
Rear mounting + usable field of view
Actual models and calibration data
Battery / rails / actuator placement
Image sharing service, if required